SORTED Postdoc Recruitment

Technical Exercises

1 Instructions

Please complete one of the following tasks: the practical task or the theoretical task. You are free to choose the task that best aligns with your interests and expertise. You will be asked to briefly present your solution to the task during the interview. This presentation may take any form you find suitable, e.g. slide deck, digital whiteboard, walking through your code, etc. You will also receive follow-up questions related to your solution, so please be prepared to discuss your approach in detail (e.g. showing code, derivations, etc).

2 The Parking Permit Problem

The parking permit problem is a classic online decision-making problem. The problem is as follows: each day, you must travel to work. You drive or cycle, depending on the weather. If you drive to work, you require a valid parking permit for that day. K different kinds of permits are available: permit of type k has duration d_k and costs c_k. A permit of duration d will allow you to drive for d consecutive days (including the day of purchase). The permit expires after d days, regardless of whether you drive or not.

3 Practical task

In this task you will implement an algorithm to solve the offline problem, i.e. determining the optimal sequence of permits to purchase given a known sequence of weather conditions. Consider the following instance of the problem: $K = 4$ types of permits are available, with durations $d = (1, 7, 14, 30)$ and costs $c = (2, 7, 12, 24)$. The weather conditions for the next 365 days are provided as an attachment to this document (see `https://umenberger.github.io/postdoc/weather.csv`). Determine the optimal sequence of permits to purchase to minimize the total cost of parking over the next 365 days. You may use any programming language and libraries you like to solve this problem.

4 Theoretical task

The parking permit problem is really an online problem, where the weather conditions are revealed one day at a time. In this task, you will analyze the performance (competitive ratio) of online mirror descent on a special case of the parking permit problem.

Assumption 4.1. You may assume the so-called interval-model of the parking permit problem: on any day t there is only one permit of each type k that covers this day. Put differently, for each permit type k, one permit covers days 1 to d_k, another permit covers days $d_k + 1$ to $2d_k$, and so on. Permits of the same type never overlap.

Assumption 4.2. You may assume that the online algorithm is allowed to purchase fractions of a permit. In reality a permit can only be purchased in whole, but allowing fractional purchases will make the analysis easier.

For some permit i, let $\mathbf{x}_i^t \in [0, 1]$ denote the fraction of permit i that is held on day t (i.e. after purchasing new permits at the beginning of day t). Let $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots)$ denote the vector that stacks the fractions of all permits. On day t, let S^t denote the set of permits that can be used to drive on day t (i.e. the permits that cover day t). If we are required to drive on day t, then the online mirror descent algorithm updates the fractions of permits held as follows:

$$\mathbf{x}^t = \arg\min_{\mathbf{x} \in \mathcal{C}^t} \; R(\mathbf{x} \mid \mathbf{x}^{t-1}) \tag{1}$$

where

$$\mathcal{C}^t = \left\{ \mathbf{x} : \sum\nolimits_{i \in S^t} \mathbf{x}_i \geq 1, \quad \mathbf{x} \geq \mathbf{x}^{t-1} \right\},$$

and $R(\mathbf{x} \mid \mathbf{x}^{t-1})$ is the regularizer given by

$$R(\mathbf{x} \mid \mathbf{y}) = \sum_i c_i \cdot D_h(\mathbf{x}_i \mid \mathbf{y}_i) \tag{2}$$

where $D_h(\mathbf{x}_i \mid \mathbf{y}_i)$ is the Bregman divergence with respect to the function $h(x) = (x + \delta) \cdot \ln(x + \delta)$ for some $\delta > 0$. Please complete the following sub-tasks:

1. If we have to drive on day t, show that the online algorithm chooses $\mathbf{x}^t$ satisfying

$$c_i \cdot \ln \left(\frac{\mathbf{x}_i^t + \delta}{\mathbf{x}_i^{t-1} + \delta} \right) = \lambda^t, \quad i \in S^t, \tag{3}$$

   where λ^t is the dual variable corresponding to the constraint $\sum_{i \in S^t} \mathbf{x}_i \geq 1$.

2. Let $\mathbf{y}$ denote the permits held by the offline optimal solution. We define the potential function

$$\Phi(\mathbf{y}, \mathbf{x}) = \epsilon \cdot \sum\nolimits_i c_i \cdot \mathbf{y}_i \cdot \ln \left(\frac{\mathbf{y}_i + \delta}{\mathbf{x}_i + \delta} \right). \tag{4}$$

   Show that when the offline optimal solutions purchases permits, the increase in potential is bounded as follows:

$$\Phi(\mathbf{y}^t, \mathbf{x}^{t-1}) - \Phi(\mathbf{y}^{t-1}, \mathbf{x}^{t-1}) \leq \epsilon \cdot \ln \left(\frac{1}{\delta} + 1 \right) \cdot \sum\nolimits_i c_i \cdot (\mathbf{y}_i^t - \mathbf{y}_i^{t-1}). \tag{5}$$

3. Show that for $\epsilon = 2$ and $\delta = (1/K)$, when the online algorithm purchases permits, the decrease in potential is bounded as follows:

$$\sum\nolimits_i c_i \cdot (\mathbf{x}_i^t - \mathbf{x}_i^{t-1}) + \Phi(\mathbf{y}^t, \mathbf{x}^t) - \Phi(\mathbf{y}^t, \mathbf{x}^{t-1}) \leq 0. \tag{6}$$

   Hint: the inequality $x - y \leq x \cdot \ln(x/y)$ for $x, y > 0$ may be useful.

4. Hence show – via a potential function argument or otherwise – that the competitive ratio of online mirror descent on the parking permit problem is at most $2 \cdot \ln(K + 1)$.